

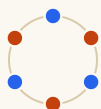
A CHAPTER IN PICTURES

The Turn That Changes Nothing

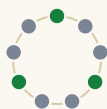
*an intuitive introduction to group theory, by way of al-Khalil's circles of Arabic
poetry*



al-Mukhtalif



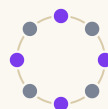
al-Mu'talif



al-Mujtalab



al-Mushtabih



al-Muttafiq

*“Symmetry, as wide or as narrow as you may define its meaning, is one idea by which
man through the ages has tried to comprehend and create order, beauty and perfection.”*

— HERMANN WEYL, *Symmetry* (1952)

§1 The Move You Cannot See

Ask a friend to close their eyes. On the ceiling above you hangs a fan with three identical blades. While they are not looking, you reach up and turn it — exactly one third of a full circle — and then you ask them to look again and say what changed.

They cannot. Not because they are careless, but because *nothing visible changed*. Each blade has moved into the position of its neighbour, and since the blades are indistinguishable, the fan as a whole looks untouched. You performed a real physical act, and the world kept no record of it.

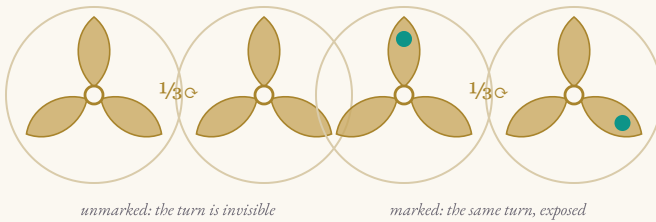


Fig. 1 — The same third-of-a-turn, twice. Unmarked (left), the move is invisible: this is what it means to be a symmetry. Mark one blade (right) and the very same move becomes plain to see.

This is the whole idea, and it is worth saying carefully:

*A **symmetry** of a thing is a move that leaves it looking exactly as it was.*

Notice what kind of definition this is. It does not say a symmetric thing is “balanced” or “even” or “pretty.” It talks about *moves* — acts you can perform — and it selects the special ones the thing cannot detect. A snowflake has the symmetries of a sixth-turn; a square has quarter-turns and flips; a human face has, roughly, one mirror. To say what symmetry something has is to list the moves it fails to notice.

That shift — from symmetry as a property you admire to symmetry as a collection of moves you perform — is the founding act of group theory. Everything in this chapter unfolds from it.

§ 1 · CONTINUED

Once you think in moves, three small observations follow — so small they feel like nothing at all.

First: doing nothing counts. Leaving the fan alone certainly leaves it looking as it was. It is the laziest symmetry, but it is one, and we will need it the way arithmetic needs zero.

Second: every move can be undone. If a third-turn clockwise leaves the fan unchanged, so does a third-turn back. Whatever a symmetry does, the reverse move is also a symmetry.

Third: two moves in a row make a move. Turn by a third, then by a third again: the result is a two-thirds turn, and the fan noticed neither. Combining symmetries can never produce anything but a symmetry — there is no way out of the collection by staying inside it.

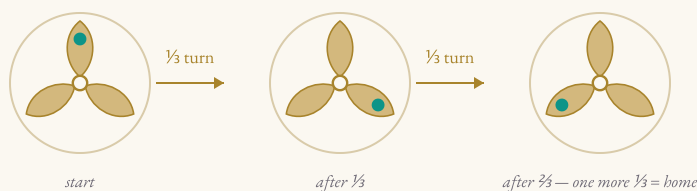


Fig. 2 — Moves compose. A third-turn followed by a third-turn is the two-thirds turn; a third-turn followed by a two-thirds turn brings every blade home — the do-nothing move.

These three remarks — a do-nothing move exists, every move reverses, moves compose — are not decoration. They are the *entire* definition of the central object of this chapter. Any collection of moves obeying them is called a **group**, and the study of such collections is group theory.

It seems impossible that so slight a definition could carry any weight. The surprise of the subject — the reason it sits at the heart of modern physics, chemistry, and cryptography — is how much follows from so little. We will watch it happen on a small stage: five circles drawn in Basra in the eighth century, to hold the metres of Arabic poetry.

§2 A Group, Properly Met

Let us be formally introduced to the fan's symmetries. There are exactly three moves it cannot detect: *stay* (do nothing), a $\frac{1}{3}$ turn, and a $\frac{2}{3}$ turn. A quarter-turn is not among them — it would leave the blades visibly askew — and a full turn is just *stay* wearing a costume.

Because two moves in a row always make a third, we can record the whole society in a little marriage-table: pick a row-move, follow it by a column-move, and read off which single move equals the pair.

then → first ↓	stay	$\frac{1}{3}$ turn	$\frac{2}{3}$ turn
stay	stay	$\frac{1}{3}$ turn	$\frac{2}{3}$ turn
$\frac{1}{3}$ turn	$\frac{1}{3}$ turn	$\frac{2}{3}$ turn	stay
$\frac{2}{3}$ turn	$\frac{2}{3}$ turn	stay	$\frac{1}{3}$ turn

Fig. 3 — The composition table of the fan's three symmetries. Every entry stays inside the collection; *stay* runs along the edges like a zero; the highlighted cells show each turn meeting its undo.

Look at what the table quietly certifies. Every combination lands back in the collection (*closure*). The *stay* row and column copy their headers, the way adding zero copies a number (*identity*). Each move finds a partner returning it to *stay*: $\frac{1}{3}$ and $\frac{2}{3}$ undo one another, and *stay* undoes itself (*inverses*). And since each entry is itself a move, chains like “ $\frac{1}{3}$ then $\frac{1}{3}$ then $\frac{2}{3}$ ” can be worked out in any grouping you please — the result is the same walk around the circle (*associativity*).

Those four certified properties are the axioms of a group — the fine print beneath the three observations of §1. This particular group, the turns of a three-fold wheel, is called the **cyclic group of order 3**, written C_3 : *cyclic* because one move (the $\frac{1}{3}$ turn), repeated, generates everything; *order 3* because there are three moves in all.

§ 2 · CONTINUED

You already operate one cyclic group every day. A clock face is a twelve-fold wheel, and “five hours after nine o’clock” is the move *advance 5* composed with the move *advance 9*: fourteen steps, which the wheel cannot tell from two. Clock arithmetic — where $9 + 5 = 2$ — is the cyclic group C_{12} doing business under an assumed name.

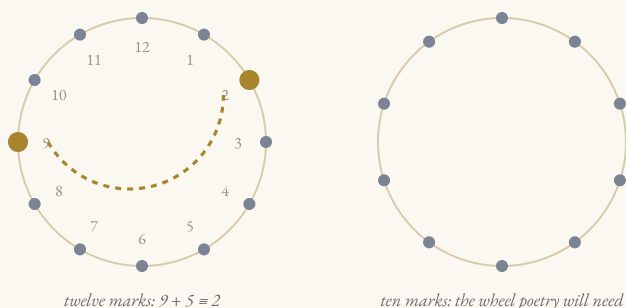


Fig. 4 — *Two wheels, one idea. On twelve marks, advancing 9 then 5 lands on 2; the wheel forgets full revolutions. Any count that wraps — hours, weekdays, drumbeats — is run by a cyclic group C_n .*

The general recipe: put n marks evenly on a wheel; the moves are *advance by k* for $k = 0, 1, \dots, n-1$; composition is addition that wraps at n . This is C_n , the cyclic group of order n — the complete symmetry inventory of anything that repeats in a loop.

Cyclic groups are the gentlest members of the group family, which is exactly why they make the right first classroom. Nothing here requires slanted algebra: a wheel, a count, a wrap. Yet they already exhibit the deep phenomena — actions, orbits, stabilizers, conservation laws — that organise far wilder groups, the ones that shuffle molecules and encrypt your messages.

And here our story proper begins. Eleven centuries before anyone wrote the word *group*, a lexicographer in Basra built five such wheels — not from brass or hours, but from the sounds of poetry — and used them to answer, completely, a question of the form group theory would later claim as its own: *how many genuinely different metres can Arabic verse possess?*

§3 The Poet's Raw Material

Arabic verse is quantitative: a line is music before it is meaning, a fixed procession of long and short. The tradition writes this rhythm with two marks. A **struck note** / is a consonant carrying a vowel — a syllable's beating heart. A **rest** 0 is an unvowelled letter that closes a syllable. Like Arabic itself, the marks are read *from the right*.

From these two sounds the prosodists built a tiny alphabet of rhythmic bricks — the pegs and cords from which every metre is assembled:

0/	0//	/0/	0///
light cord	joined peg	split peg	little cluster
سبب خفيف	وتد مجموع	وتد مفروق	فاصلة صغرى
strike, rest — a light two-step	strike, strike, rest — the spine of most metres	strike, rest, strike — rare, and fateful in §8	three strikes, rest — a quick drum-roll

Fig. 5 — The four atomic units of Arabic prosody. Read each symbol from the right: 0// is strike–strike–rest. The colours follow these units through every figure of this chapter.

Al-Khalīl ibn Aḥmad al-Farāhīdī (الخليل بن أحمد الفراهيدي)

Born in Oman around 718 CE, working in Basra until his death around 786, al-Khalīl compiled the first dictionary of Arabic (*Kitāb al-'Ayn*), organised by where in the mouth each sound is made; taught Sibawayh, author of the foundational Arabic grammar; and single-handedly founded *'ilm al-'arūd*, the science of metre. Tradition says the system came to him from the rhythm of a coppersmith's hammer in the market. What he did with it, the next pages unfold.

§ 3 · CONTINUED

Take the most beloved metre of classical poetry, الطويل (*aṭ-Ṭawīl*, “the long one”). Its half-line is the chant فَعُولُنْ مَفَاعِيلُنْ فَعُولُنْ مَفَاعِيلُنْ — in our marks, the ten-unit procession 0// 0/ 0// 0/ 0/ repeated twice. Verse after verse, line after line, the same procession returns. The pattern is a loop pretending to be a line.

Al-Khalil’s decisive act was to stop pretending: *roll it up*. Bend the repeating tape until its end meets its beginning, and the metre becomes a necklace of ten beads.

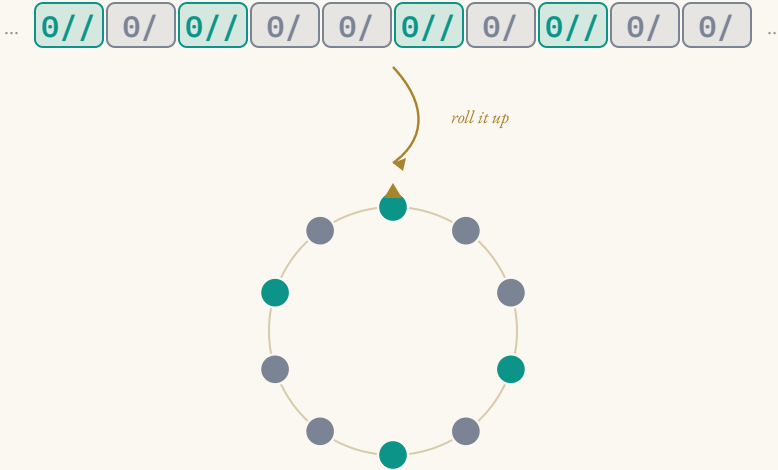


Fig. 6 — Rolling the repeating line into a circle. The tape’s endless repetition becomes a single loop of ten units — teal joined pegs, grey light cords. A mathematician calls this a *quotient*: the infinite line divided by its own repetition.

This rolling-up is itself a mathematical construction, and a deep one. Whenever a pattern repeats every n steps, positions that differ by n are “the same,” and gluing all the same together turns the infinite line of integers into a wheel with n marks — the very wheel of Fig. 4. The circle is not a picture of the metre; it *is* the metre, with the bookkeeping of repetition thrown away.

Five such necklaces, al-Khalil found, suffice for all of Arabic poetry. The first — the one above, called دائرة المختلف, *the circle of the differing* — will be our specimen. It has ten beads: three teal pegs, seven grey cords. And a wheel with ten marks has, waiting in the wings, a group.

§4 One Necklace, Many Clasps

A necklace has no beginning. To recite a metre, though, you must start somewhere — you need to place a *clasp*, an opening point, and read the beads from there around the loop. And now the group arrives on stage: C_{10} , the ten rotations of a ten-bead wheel, *acts* on the necklace. Each rotation carries the clasp to a new bead.

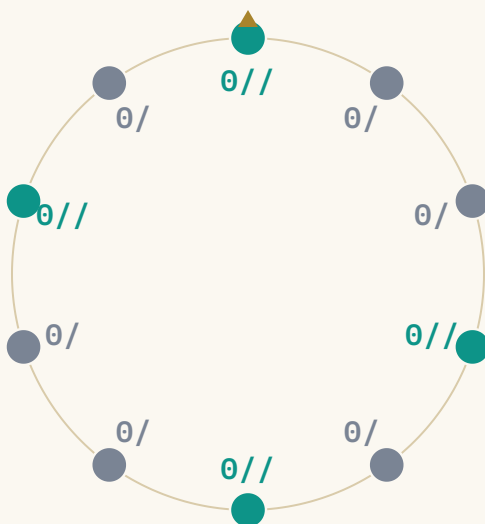


Fig. 7 — The circle of the differing (دائرة المختلف), beads labelled and coloured, read counter-clockwise from the marked clasp at the top. Reading from the top bead yields فعولن مضاعين ... — the metre at-Ṭawīl.

Here is the beautiful part. Move the clasp one bead onward and read again: the same beads, in the same cyclic order, deliver a *different metre* — different feet, a different music, a different name in the anthologies. The rotation did not merely relabel the necklace; it changed which poem you are hearing.

When a group's moves are applied to an object this way, mathematicians say the group *acts* on the object, and each move-result is a fresh point of view on the same underlying thing. Al-Khalīl's insight, translated forward eleven centuries, is exactly this: *the metres of Arabic poetry are not sixteen separate inventions. They are a handful of necklaces, viewed through the rotations of cyclic groups.*

So — ten rotations, ten possible clasps. Do we get ten metres from this circle? Turn the page; the answer is the first genuine surprise, and the hinge of the whole chapter.

§ 4 · CONTINUED

Let us do it honestly: place the clasp at each of the ten beads in turn, read off the pattern, and check it against the poets' repertoire.

0	0//	0/	0//	0/	0/	0//	0/	0//	0/	0/	الطويل · al-Ṭawīl, “the long”
1	0/	0//	0/	0/	0//	0/	0//	0/	0/	0//	المديد · al-Madīd, “the protracted”
2	0//	0/	0/	0//	0/	0//	0/	0/	0//	0/	المستطيل · al-Mustaṭīl — possible, unused
3	0/	0/	0//	0/	0//	0/	0/	0//	0/	0//	البسيط · al-Basīṭ, “the outspread”
4	0/	0//	0/	0//	0/	0/	0//	0/	0//	0/	الممتد · al-Mumtadd — possible, unused
5– 9	the same five readings again, in the same order — nothing new appears										

Fig. 8 — All ten clasp positions of the first circle. Five distinct readings emerge — three canonical metres and two lawful strangers — and then the list repeats itself.

Two things on this list demand explanation, and each gets its own section.

The list is short. Ten rotations produced only five readings: clasp position 5 sounds exactly like position 0, position 6 like 1, and so on. Some rotations changed nothing at all. Why five, and not ten or seven? That is §5 and §6 — orbits and stabilizers.

The list is too long. The poets use three of these five: البسيط، المديد، الطويل. The other two are perfectly well-formed rhythms that classical poetry simply never adopted. Al-Khalīl did not discard them; he named them and set them aside. Their story — the empty cells of the catalogue — is §9.

§5 The Orbit: What Is Genuinely Different

Collect the distinct readings a necklace offers — one for each *genuinely different* clasp position — and you have what group theory calls the **orbit** of the necklace under rotation: every place the object can be carried by the group’s moves, counted without repetition.

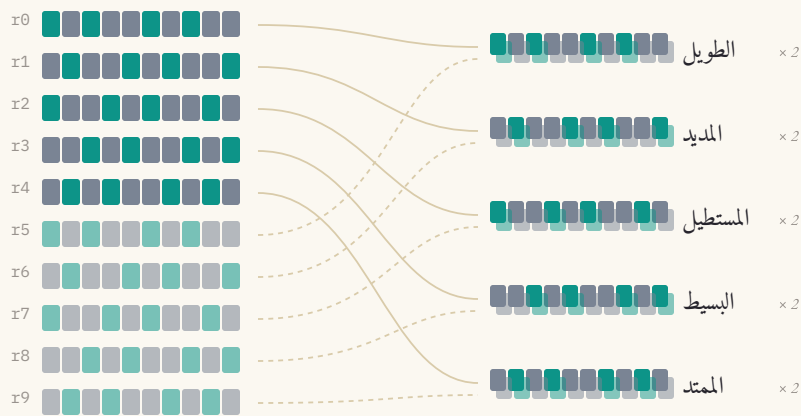


Fig. 9 — The ten rotations of circle 1, dealt into piles that sound identical. Ten readings collapse into five piles of exactly two. The orbit has five members; the pile-height — here, two — will get its own name overleaf.

The orbit is the honest inventory. The group has ten moves, but moves are labour, not merchandise: what the necklace actually *yields* is five rhythms. In the same spirit, all five of al-Khalil’s circles can be audited. The bead-counts differ, the units differ, and the yields differ strikingly:

CIRCLE	BEADS N	ORBIT (DISTINCT READINGS)
المختلف · the differing	10	5
المؤتلف · the concordant	6	2
المجتلب · the brought-in	9	3
المشتبه · the resembling	9	9
المتفق · the agreeing	8	2

Fig. 10 — Five necklaces, audited. The orbits sum to $5+2+3+9+2 = 21$ possible readings in the whole system. Note the two nine-bead circles with wildly different yields: 3 against 9. Explaining that is where the theory earns its keep.

§6 The Stabilizer: The Necklace's Own Symmetry

Why did clasp position 5 sound exactly like position 0? Look at the first circle's beads: 0// 0// 0// 0// 0// — and then the same five again. The sequence is built of two identical halves. So the rotation by five beads carries every peg onto a peg and every cord onto a cord: it is a move the *necklace itself* cannot detect. The fan of §1 has returned, wearing beads.



Fig. 11 — The turn that changes nothing, performed. A ghost copy of the necklace rotates by five positions and settles exactly onto the original — peg on peg, cord on cord.

The moves that leave an object looking untouched form a little group of their own inside the big one (compose two undetectable moves, or undo one, and the object still detects nothing). This inner circle is called the **stabilizer** of the object — its private symmetry group. For circle 1 the stabilizer is { stay, turn-by-5 }: a copy of C_2 , two moves strong.

And now the shortness of Fig. 8's list explains itself. Start from any clasp position and apply a stabilizer move: the reading cannot change — that is what “undetectable” means. Ten clasp positions, sorted two-and-two by the stabilizer, give five genuinely different readings. The orbit is small precisely *because* the necklace is symmetric.

§ 6 · CONTINUED

A symmetry can be *drawn*. On each circle, join every bead to the bead its stabilizer-turn sends it to. The chords trace a star polygon — and because an undetectable move must send each bead to a bead of the same kind, every chord joins two beads of one colour.

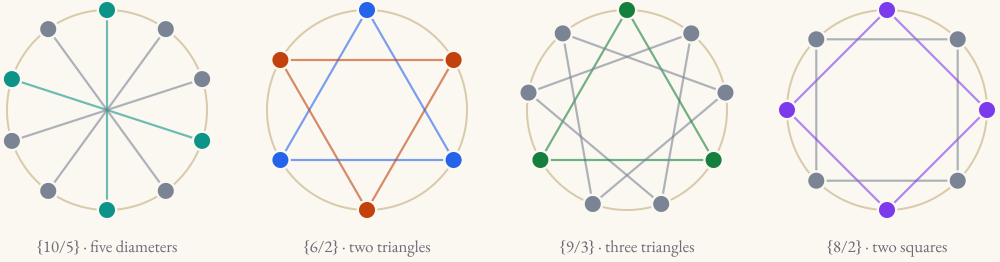


Fig. 12 — Four circles, their stabilizers drawn as stars. In the concordant circle the hexagram splits into a blue peg-triangle and a rust cluster-triangle: the symmetry literally sorts the beads by kind. One circle is missing from this gallery — it draws no star at all, and it is the subject of §8.

The shepherd's theorem

A shepherd counts forty legs and announces ten sheep. It works because every sheep contributes *exactly the same number* of legs: divide the easy count by the per-item redundancy to get the honest one. Keep this trick in hand; on the next page it becomes, almost word for word, a theorem of group theory.

One habit of mind before moving on: the stabilizer measures *self*-symmetry. The group C_{10} belongs to the wheel — to the ten positions. The stabilizer belongs to the *beading*: thread the same wheel differently and its stabilizer changes. Symmetry is not in the stage; it is in what stands on it.

§7 A Conservation Law

We have counted three different things on the first circle: the group’s moves (ten), the orbit’s readings (five), the stabilizer’s undetectable turns (two). Set them side by side:

$$10 = 5 \times 2$$

$$\text{moves} = \text{distinct readings} \times \text{symmetries}$$

This is no coincidence of the number ten, and the shepherd already told us why. Deal the ten rotations into piles according to *which reading they produce*. Every pile has exactly the stabilizer’s size: if two rotations give the same reading, they differ by an undetectable move, and each reading’s pile is just the stabilizer shifted over — same size, different sheep.

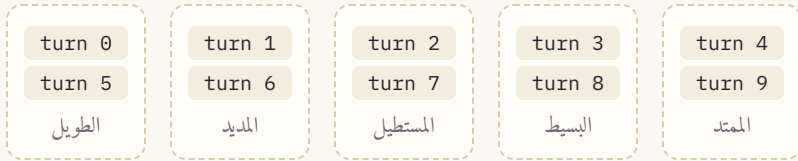


Fig. 13 — The ten moves of C_{10} dealt by the reading they produce. Five piles, each of height exactly two — the stabilizer’s size. Equal piles are the whole proof: $10 = 5 \text{ piles} \times 2 \text{ per pile}$.

Stated in full generality, this is the **orbit–stabilizer theorem**: for a finite group acting on an object,

$$\text{size of group} = \text{size of orbit} \times \text{size of stabilizer}.$$

It has the shape of a conservation law. A necklace cannot be both rich in symmetry and rich in readings: their product is pinned to the size of the group, so whatever one gains, the other must surrender. Variety and symmetry are two spendings of a single budget.

§7 · CONTINUED

A law is only as good as its audits. Here are all five circles, each row an independent test the eighth-century data must pass:

CIRCLE	BEADS	ORBIT	STABILIZER	THE LAW
المختلف	10	5	C_2	$10 = 5 \times 2$
المؤتلف	6	2	C_3	$6 = 2 \times 3$
المجتلب	9	3	C_3	$9 = 3 \times 3$
المشتبه	9	9	C_1	$9 = 9 \times 1$
المتفق	8	2	C_4	$8 = 2 \times 4$

Fig. 14 — *One theorem, five costumes. Every circle of the classical system satisfies beads = orbit $\times$ stabilizer exactly — no slack, no exception. (All counts in this chapter are verified by exhaustive computation; see the colophon.)*

Read the table twice, in opposite directions. Downward, it is bookkeeping: five instances of a multiplication. Across, it is a see-saw. The concordant and agreeing circles carry the largest stabilizers in the system — C_3 on six beads, C_4 on eight — and are duly the poorest, two readings apiece. The resembling circle carries no symmetry at all and is the richest: all nine of its rotations sound distinct.

Hold on to the two middle rows. Nine beads twice over — yet one nine-bead necklace yields three readings and the other yields nine. Same wheel, same group C_9 , same bead-count; only the *threading* differs. Whatever separates those two rows is not arithmetic but structure, and it is the finest story in the whole system. It is next.

§8 Symmetry Comes in Steps, Not Slopes

Before comparing the two nine-bead circles, one more gift from the equal piles. The stabilizer's size always divides the group's size exactly — piles of equal height must share the total with no remainder.

(This is **Lagrange's theorem**, and it governs every subgroup of every finite group.) The consequence deserves a slow reading: *the sizes of possible symmetry are not a dial; they are a short ladder of divisors.*



Fig. 15 — The only symmetries a necklace may have, by Lagrange: divisor ladders. On nine beads nothing exists between C_3 and C_1 — there is no such thing as “almost” threefold symmetry. (The top rung, all beads identical, occurs in no real circle.)

On nine beads, then, the choices are stark: full ninefold sameness (impossible unless every bead is identical), threefold symmetry, or none. No seven-eighths symmetry, no gentle degradation. A nine-bead necklace that loses its C_3 does not slip to something slightly weaker — *there is nothing slightly weaker*. It falls the whole way to C_1 , the trivial group whose only member is *stay*.

Which is exactly the cliff-edge on which the two nine-bead circles stand, one on each side.

§ 8 · CONTINUED

Lay the two nine-bead circles out flat and compare, bead by bead:

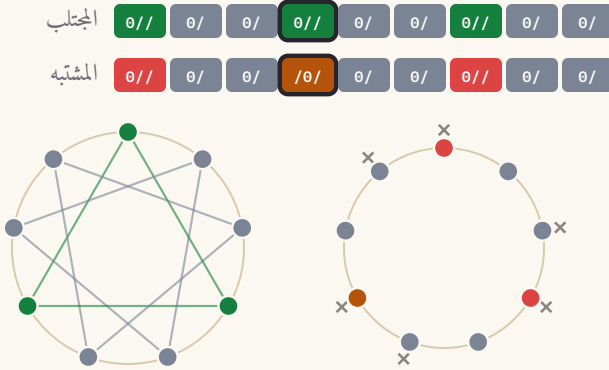


Fig. 16 — One bead of difference. The brought-in circle repeats peg–cord–cord three times: symmetry C_3 , three readings. The resembling circle replaces the middle peg with the rare split peg $/0/$ — and the smallest turn now sets six beads on unlike beads (the $\times$ marks). Symmetry: none. Readings: all nine.

One bead out of nine — the least possible alteration — and by the ladder of Fig. 15 the collapse is total: C_3 to C_1 , orbit three to orbit nine. The most similar pair of circles in the system sit at the greatest possible distance in symmetry. Al-Khalil’s tradition named the second circle *المشبه* — *the resembling*, and in the same breath *the confusable*, for in Arabic the two are one word: things that resemble each other closely enough defeat telling-apart. Its rotations are near-misses of one another, and its six metres are precisely the ones students of prosody perennially mix up.

There is a quiet lesson here about invariants. The stabilizer records *that* symmetry is absent — it says C_1 , exactly as it would for random noise. It cannot say *how narrowly* symmetry was missed; the group forgets the nearness. The old name remembers what the invariant forgets. A good measurement, one might say, and a better ear.

§9 The Catalogue and Its Gaps

Add the orbits: $5 + 2 + 3 + 9 + 2 = 21$. The five circles offer twenty-one genuinely different readings. The poets, across the whole classical corpus, used sixteen. Al-Khalil saw both numbers — and refused to trim his theory to fit the evidence.

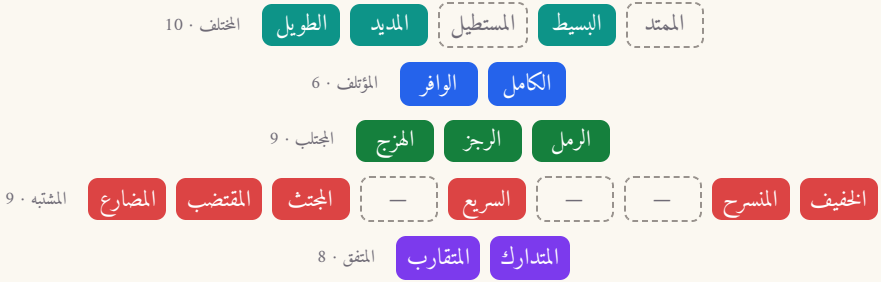


Fig. 17 — The complete theoretical catalogue: 21 cells, one per distinct reading, row by circle. Sixteen are inhabited by the classical metres. Five are empty (dashed) — lawful rhythms no poet adopted. Al-Khalil named the first circle's two vacancies; the resembling circle's three were left nameless.

Those dashed cells are the most modern thing in this chapter. A catalogue built from evidence alone lists what occurs. A catalogue built from a *group action* lists what is possible — the orbit does not care what happens to be attested. The five vacancies are predictions: here are rhythms the language permits, whether or not anyone has sung them. And absence itself becomes a fact in need of explanation — metrical, aesthetic, historical — where before there was nothing to explain.

Eleven centuries later, Mendeleev would do exactly this with the elements: trust the structure, print the empty boxes, and let gallium and germanium arrive to fill them. Al-Khalil's empty boxes are still empty — poets are not obliged to oblige — but later experimenters did try the vacant metres, and *المستطيل* and *الممتد* live on as curiosities in the manuals, wearing the names he gave to rhythms nobody used. Note, too, *where* the gaps fall: all five sit in the two least symmetric circles, the only ones whose orbits offered more than the tradition cared to take.

§10 The Price of Symmetry

One law has been running beneath every page, and it deserves to be seen whole. The orbit–stabilizer theorem fixes the product *orbit* $\times$ *stabilizer*; so across the five circles, symmetry and variety must trade against each other exactly:

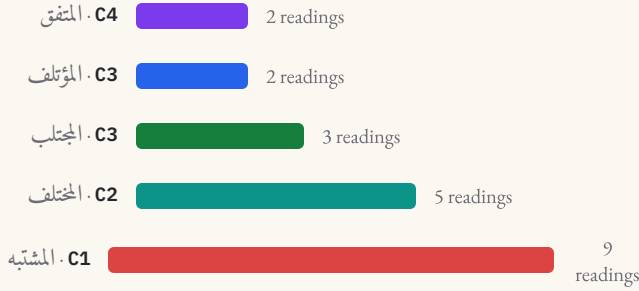


Fig. 18 — *The inverse law, drawn. Circles sorted by symmetry, from the most symmetric (C_4) to the least (C_1); the yield of distinct readings runs the other way. The most symmetric circle is the poorest; the one with no symmetry at all is the richest in the system.*

Everyday speech treats symmetry as wealth — the more, the better. Group theory’s ledger reads the opposite way: *symmetry is sameness, and sameness collapses possibility*. Every undetectable turn a necklace keeps is a clasp position it forfeits; perfect symmetry would mean a single reading, perfect monotony. It is the breaking of symmetry that creates distinct things.

Physics tells the same story at other scales — the early universe cooling through broken symmetries into the zoo of particles; a crystal’s perfect lattice hosting one pattern where a disordered glass hosts endlessly many. The circles of al-Khalil tell it in nine beads: the concordant circle, most symmetric, sings two songs; the resembling circle, symmetric not at all, sings nine — and confuses everyone.

§ 10 · CLOSING

Step back and count what has been assembled, because it is more than a stack of definitions. A collection of objects (rhythms); a group of moves acting on them (rotations); a rule declaring two objects *the same* when a move joins them; the same-classes tallied (orbits — the metres); each object's private symmetry measured (stabilizers); a conservation law binding the counts; an exhaustive catalogue derived rather than collected; and vacancies in the catalogue treated as predictions. That list is a complete description of the classification method that carries modern mathematics — and every item on it was executed, by ear, in eighth-century Basra, eleven hundred years before Galois gave the subject its name in the 1830s.

This is the deepest thing the circles teach about group theory: that it is not, at bottom, a notation. It is a discipline for a question every careful practice eventually asks — *when are two things genuinely the same, and how many genuinely different things are there?* Ask it honestly enough, of molecules or ciphers or verse, and orbits and stabilizers assemble themselves in your hands, whether or not anyone has told you their names. The formalism did not invent the symmetry of poetry. It arrived late, and politely gave a name to what a lexicographer had already heard.

Hear it for yourself

Every figure in this chapter is a still from a living instrument: the *Interactive Arud Explorer* at arudi.midadalfikr.com. There you can turn each circle's dial with a finger and watch metres change at every détente; perform the ghost-rotation of Fig. 11 and see the stars of Fig. 12 draw themselves; collapse the ten rotations into their five piles; and — best of all — press play, and *listen* to a necklace as a drum pattern, moving the clasp while it sounds. The mathematics of this chapter is entirely contained in what your ear will notice.

And the next time you turn something and no one can tell — a fan, a bolt, a phrase — remember that you have performed an element of a stabilizer, and that a poet could have told you which one.

Glossary & Notes

symmetry

a move that leaves a thing looking exactly as it was (§1).

group

a collection of moves containing *do-nothing*, closed under composition, with every move undoable (§2).

cyclic group C_n

the n rotations of an n -mark wheel; clock arithmetic (§2).

action

a group applied to an object, each move giving a new view of it (§4).

orbit

everything an object becomes under the group's moves, counted without repetition — here, the distinct metres of a circle (§5).

stabilizer

the moves an object cannot detect; its private symmetry group (§6).

orbit–stabilizer theorem

group size = orbit $\times$ stabilizer (§7).

Lagrange's theorem

a subgroup's size divides the group's size; symmetry comes in divisor steps (§8).

quotient

gluing together positions declared “the same” — rolling the repeating line into the circle (§3).

عروض *arūd*

the science of Arabic metre, founded by al-Khalil (§3).

sabab سبب, watid وتد, fāṣila فاصلة

cord, peg, and cluster — the atomic rhythm units (Fig. 5).

muhmal مَهْمَل

“neglected”: a lawful reading no poet adopted — the catalogue's empty cells (§9).

the five circles

اختلف the differing (10 beads), المؤتلف the concordant (6), المتقلب the brought-in (9), المشتبّه the resembling (9), المتفق the agreeing (8).

the sixteen metres

الطويل، المديد، البسيط · الوافر، الكامل · الخرج، الرجز، الرمل · السريع، المنسرح، الخفيف، المضارع، المقتضب، المجتث · المتقارب، المتدارك.

FURTHER READING. H. Weyl, *Symmetry* (1952) — the classic meditation; M. A. Armstrong, *Groups and Symmetry* (1988) — a first formal course; and any classical manual of *ʿilm al-ʿarūd* for the circles as the tradition drew them.

COLOPHON. Set in EB Garamond, Cormorant Garamond, Amiri, and IBM Plex Mono. All counts used in this chapter — orbits 5, 2, 3, 9, 2; stabilizers C_2 , C_3 , C_3 , C_1 , C_4 ; $21 = 16 + 5$ — were verified by exhaustive computation against the sequence data of the Interactive Arud Explorer. Prosodic symbols appear in classical right-to-left order throughout: 0// is read strike–strike–rest, from the right.